10.18

(a)

```
母親大學教育比例: 12.14953 %
父親大學教育比例: 11.68224 %
```

(b)

```
[1] "相關係數矩陣:"
                educ MOTHERCOLL FATHERCOLL
educ       1.0000000  0.3594705  0.3984962
MOTHERCOLL 0.3594705  1.0000000  0.3545709
FATHERCOLL 0.3984962  0.3545709  1.0000000
```

使用大學學歷虛擬變數（Dummy）而非教育年數作為工具變數，通常是因為「大學學位」具有強烈的門檻效應，能過濾掉部分測量誤差，且在社會學上，父母是否完成大學學業對子女的激勵作用往往比單純多讀一年書更顯著。

(c)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
    exper + I(exper^2), data = mroz_working)

Residuals:
     Min       1Q   Median       3Q      Max
-3.08719 -0.32444  0.04147  0.36634  2.35621

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.1327561  0.4965325  -0.267  0.78932
educ         0.0760180  0.0394077   1.929  0.05440 .
exper        0.0433444  0.0134135   3.231  0.00133 **
I(exper^2)  -0.0008711  0.0004017  -2.169  0.03066 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared: 0.147,      Adjusted R-squared: 0.1409
Wald test:   8.2 on 3 and 424 DF,  p-value: 2.569e-05

            2.5 %    97.5 %
educ -0.001219763 0.1532557
```

雖然係數為正，但 p 值為 0.0544，在 5% 顯著水準下剛好不顯著。

(d)

```
Call:
lm(formula = educ ~ MOTHERCOLL + exper + I(exper^2), data = mroz_working)

Residuals:
    Min      1Q  Median      3Q     Max 
-7.4267 -0.4826 -0.3731  1.0000  4.9353 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 12.079094   0.303118  39.849  < 2e-16 ***
MOTHERCOLL   2.517068   0.315713   7.973 1.46e-14 ***
exper        0.056230   0.042101   1.336    0.182    
I(exper^2)  -0.001956   0.001256  -1.557    0.120    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.133 on 424 degrees of freedom
Multiple R-squared:  0.1347,	Adjusted R-squared:  0.1285 
F-statistic: 21.99 on 3 and 424 DF,  p-value: 2.965e-13


Linear hypothesis test:
MOTHERCOLL = 0

Model 1: restricted model
Model 2: educ ~ MOTHERCOLL + exper + I(exper^2)

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)    
1    425 2219.2                                  
2    424 1929.9  1    289.32 63.563 1.455e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

由於 F 值遠大於經驗法則的 10，MOTHERCOLL 是一個強工具變數，與教育高度相關。

(e)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL + 
    FATHERCOLL + exper + I(exper^2), data = mroz_working)

Residuals:
     Min       1Q   Median       3Q      Max 
-3.07797 -0.32128  0.03418  0.37648  2.36183 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)   
(Intercept) -0.2790819  0.3922213  -0.712  0.47714   
educ         0.0878477  0.0307808   2.854  0.00453 **
exper        0.0426761  0.0132950   3.210  0.00143 **
I(exper^2)  -0.0008486  0.0003976  -2.135  0.03337 * 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,	Adjusted R-squared: 0.147 
Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06 

          2.5 %    97.5 %
educ 0.02751845 0.1481769
```

此區間比（c）窄。這符合理論：增加有效的工具變數能提高估計的精確度，並使教育對薪資的正向影響在 5% 水準下變得顯著（p = 0.0045）。

(f)

```
Call:
lm(formula = educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2),
    data = mroz_working)

Residuals:
    Min      1Q  Median      3Q     Max 
-7.2152 -0.3056 -0.2152  0.7627  5.0620 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 11.890259   0.290251  40.965  < 2e-16 ***
MOTHERCOLL   1.749947   0.322347   5.429 9.58e-08 ***
FATHERCOLL   2.186612   0.329917   6.628 1.04e-10 ***
exper        0.049149   0.040133   1.225    0.221    
I(exper^2)  -0.001449   0.001199  -1.209    0.227    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.033 on 423 degrees of freedom
Multiple R-squared:  0.2161,	Adjusted R-squared:  0.2086 
F-statistic: 29.15 on 4 and 423 DF,  p-value: < 2.2e-16


Linear hypothesis test:
MOTHERCOLL = 0
FATHERCOLL = 0

Model 1: restricted model
Model 2: educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2)

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)    
1    425 2219.2                                  
2    423 1748.3  2    470.88 56.963 < 2.2e-16 ***
```

- 第一階段 F 值：56.963，依然非常強勁。
- Wu-Hausman 檢定：p = 0.472，無法拒絕教育為外生的假設（但在 IV 框架下我們通常仍採用 IV 估計以應對潛在內生性）。
- Sargan 檢定：p = 0.626，無法拒絕 $H_0$，代表工具變數滿足外生性條件，沒有過度識別的問題。

(g)

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
    FATHERCOLL + exper + I(exper^2), data = mroz_working)

Residuals:
     Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819  0.3922213  -0.712  0.47714
educ         0.0878477  0.0307808   2.854  0.00453 **
exper        0.0426761  0.0132950   3.210  0.00143 **
I(exper^2)  -0.0008486  0.0003976  -2.135  0.03337 *

Diagnostic tests:
                 df1 df2 statistic p-value
Weak instruments   2 423    56.963  <2e-16 ***
Wu-Hausman         1 423     0.519   0.472
Sargan             1  NA     0.238   0.626
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,      Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06
```

虛無假設:所有的工具變數都是外生的(即 Mothercoll 與 Fathercoll 皆為有效工具變數，且與誤差項不相關

P value 大於 0.05，無法拒絕虛無假設。代表這兩個工具變數是有效的，多加入的工具變數與工資方程式的誤差項沒有顯著相關，因此模型設定是正確的。

10.20

(a)

```
Call:
lm(formula = msft_premium ~ mkt_premium, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max
-0.27424 -0.04744 -0.00820  0.03869  0.35801

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.003250   0.006036   0.538    0.591
mkt_premium 1.201840   0.122152   9.839   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523,    Adjusted R-squared:  0.3486
F-statistic:  96.8 on 1 and 178 DF,  p-value: < 2.2e-16
```

由於 beta > 1，微軟被歸類為冒險性 (Risky) 股票，其波動程度高於市場平均水平。

(b)

```
Call:
lm(formula = mkt_premium ~ RANK, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.903e-02  2.195e-03   -36.0   <2e-16 ***
RANK         9.067e-04  2.104e-05    43.1   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,	Adjusted R-squared:  0.9121
F-statistic:  1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

RANK 與市場溢酬高度相關，是一個極強的工具變數。

(c)

```
Call:
lm(formula = msft_premium ~ mkt_premium + v_hat, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005984   0.504   0.6146
mkt_premium  1.278318   0.126749  10.085   <2e-16 ***
v_hat       -0.874599   0.428626  -2.040   0.0428 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,	Adjusted R-squared:   0.36
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

在 5% 顯著水準下，v_hat 是顯著的，說明市場溢酬存在內生性問題（測量誤差），但在 1% 水準下則不顯著。

(d)

```
Call:
ivreg(formula = msft_premium ~ mkt_premium | RANK, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max
-0.271625 -0.049675 -0.009693  0.037683  0.355579

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.003018   0.006044   0.499    0.618
mkt_premium 1.278318   0.128011   9.986   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared: 0.3508,	Adjusted R-squared: 0.3472
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16
```

此值高於 OLS 的 1.2018。這符合衰減偏誤 (Attenuation Bias) 的預期：當

自變數有測量誤差時，OLS 會低估係數，而 IV 能將其修正回來。

(e)

```
Call:
lm(formula = mkt_premium ~ RANK + POS, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.109182 -0.006732  0.002858  0.008936  0.026652 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.0804216  0.0022622  -35.55   <2e-16 ***
RANK         0.0009819  0.0000400   24.55   <2e-16 ***
POS         -0.0092762  0.0042156   -2.20   0.0291 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,	Adjusted R-squared:  0.9139 
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16


Linear hypothesis test:
RANK = 0
POS = 0

Model 1: restricted model
Model 2: mkt_premium ~ RANK + POS

  Res.Df     RSS Df Sum of Sq      F    Pr(>F)    
1    179 0.43784                                  
2    177 0.03727  2   0.40057 951.26 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

RANK 與 POS 共同構成強大的工具變數組合。第一階段 $R^2$ 微幅上升至 0.9149。

(f)

```
Call:
lm(formula = msft_premium ~ mkt_premium + v_hat_e, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27132 -0.04261 -0.00812  0.03343  0.34867 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.003004   0.005972   0.503   0.6157    
mkt_premium  1.283118   0.126344  10.156   <2e-16 ***
v_hat_e     -0.954918   0.433062  -2.205   0.0287 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,	Adjusted R-squared:  0.3625 
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

在 1% 顯著水準下，我們仍**無法**得出市場報酬是內生的結論（因為 0.0287 > 0.01），但在 5% 水準下則顯著。

(g)

```
Call:
ivreg(formula = msft_premium ~ mkt_premium | RANK + POS, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27168 -0.04960 -0.00983  0.03762  0.35543 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003004   0.006044   0.497     0.62    
mkt_premium 1.283118   0.127866  10.035   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared: 0.3507,	Adjusted R-squared: 0.347 
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16 
```

估計值進一步增加。這加強了「OLS 因測量誤差而低估微軟風險」的觀點。IV 估計出的 Beta 更能反映微軟與市場真實的敏感度關係。

(h)

```
LM Statistic: 0.5584634 
P-value: 0.45488 
```

P-value $< 0.05$，則拒絕虛無假設，表示工具變數可能無效。